Harsh Sinha

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Education

University of Pittsburgh 2021 - present
Ph.D. Intelligent Systems *GPA :* 3.7/4.0
Thesis : From Quality Control to Biological Discovery:
Computational Methods for Spatial Transcriptomics

Birla Institute of Technology and Science, Pilani 2018 - 20
M.E. Software Systems *GPA :* 7.96/10.0
Thesis : Hyperspectral guided Image Dehazing

Birla Institute of Technology and Science, Pilani 2013 - 18
M.Sc. (Hons.) Chemistry *GPA :* 7.96/10.0
Thesis : Emission Properties of triplet harvesting AIE active materials in
the reverse micelles of gemini surfactants: Effect of spacer chain length

Birla Institute of Technology and Science, Pilani 2013 - 18
B.E. (Hons.) Electrical & Electronics *GPA :* 7.96/10.0
Thesis : Deep Learning for Biometrics

Publications

Unifying AI-assisted scientific discovery around exploration, hypothesis generation, and testing
MM Hasib, S Jo, H Sinha, J Song, A Das, Z Liu, H Galloway, H Huang, JM Zhang, K Zhang, SJ Gao, YC Chiu, L Li, and Y Huang
In Patterns, 101660, 2026

STiLE: Automated Tissue Microarray Dearraying for Spatial Transcriptomics
H Sinha, A Das, YC Chiu, SJ Gao, Y Huang
In Bioinformatics, 42(10), btag572, 2026

Solving the Pervasive Problem of Protocol Non-Compliance in MRI using an Open-Source tool *mrQA*
H Sinha and PR Raamana
In Neuroinformatics, 2024
In Alzheimer's Association International Conference (AAIC) 2024
In Organization for Human Brain Mapping Annual Meeting (OHBM) 2023 & 2024
In ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM) 2023 & 2024

***MRdataset*: A unified and user-friendly interface to medical imaging datasets**
H Sinha and PR Raamana
In Journal of Open Source Software, 9(95), 6269, 2024
In INCF Neuroinformatics Assembly 2023

Towards Shape-regularized Learning for Mitigating Texture Bias in CNNs
H Sinha and A Kovashka
In ACM International Conference on Multimedia Retrieval (ICMR 2023) | Oral Session

Cycle Consistent Embedding of 3D Brains with Auto-Encoding Generative Adversarial Networks
S Xing, H Sinha, and SJ Hwang
In International Conference on Medical Imaging with Deep Learning (MIDL 2021)

Learning to Enhance Visual Quality via Hyperspectral Domain Mapping
H Sinha, A Mehta, M Mandal and P Narang
In AAAI Conference on Artificial Intelligence (AAAI 2021, Student Abstract)

Domain-Aware Unsupervised Hyperspectral Reconstruction for Aerial Image Dehazing
A Mehta, H Sinha, M Mandal and P Narang
In IEEE Winter Conference on Applications of Computer Vision (WACV 2021)

HIDeGAN: A Hyperspectral-guided Image Dehazing GAN
A Mehta, H Sinha, P Narang and M Mandal
In IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2020)

Audio Classification using Braided Convolutional Neural Networks
H Sinha, V Awasthi and Pawan K Ajmera
In IET Signal Processing, 2020

Text-Convolutional Neural Networks for Fake News Detection in Tweets.
H Sinha, S Kalra and Y Sharma.
In Evolution in Computational Intelligence, Springer, 2020.

Upgrading Security and Protection in Ear Biometrics
H Sinha and PK Ajmera
In IET Biometrics, 2019

Convolutional Neural Network-based Human Identification using Outer Ear Images
H Sinha,R Manekar, Y Sinha, and Pawan K Ajmera
In Soft Computing for Problem Solving, 2019

Studying the Role of Kinect as a Multi-Sensory Learning Platform for Children
H Sinha, S Srivastava, and Y Sinha
In ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp 2018)

Interweaving Convolutions: An application to Audio Classification
H Sinha and PK Ajmera
In International Conference on Knowledge Discovery and Data Mining, DL Day Workshop (KDD 2018)

Talks & Presentations

Alzheimer's Association International Conference (AAIC)

2024 | Indiscriminate Adjustment for Confounders is worse than you think and What can be done about it
2023 | Bridging the Brain-Age Gap: Quantifying regional contributions using Shapley-Owen values

INCF Neuroinformatics Assembly

2023 | Building bridges in MR research: unified interface to different MRI dataset formats with MRdataset
2022 | Garbage in, Garbage out: Protocol Compliance, Integrity and Quality Assurance of Neuroimaging Datasets

Teaching

University of Pittsburgh

Discrete Structures for CS	Fall 2026, Fall 2023
Introduction to Computing for the Humanities	Fall 2026, Fall 2023
Introduction to Computing for Scientists	Fall 2026
Introduction to Machine Learning	Spring 2024

Johns Hopkins University

Exploratory Data Analysis	Summer 2015
Reproducible Research	Summer 2015

Experience

CSIR-CEERI, Pilani July–Dec 2018
Mapped 3-channel RGB images to 31-band hyperspectral images using GANs; evaluated reconstruction quality with SSIM and PSNR.

Samsung R&D Institute, Noida Summer 2017
Built an end-to-end encoder-decoder for pixel-level blur detection without external estimation; integrated the trained model into an Android app using TensorFlow Lite.

Honors & Awards

ISMRM & ISMRT Annual Meeting & Exhibition Educational Stipend	2023, 2024
Intelligent Systems Program Fellowship	2021-2023
UbiComp 2018 Travel Grant	Oct 2018
Microsoft Code.Fun.Do. Finalist Forum	Feb 2016 - Aug 2016

Service

ACM Web Conference (WWW)	2027
IEEE International Conference on Robotics and Automation (ICRA)	2027
Discover Artificial Intelligence	2026
Physical and Engineering Sciences in Medicine	2026
International Conference of IEEE Engineering in Medicine and Biology and Society (EMBC)	2025
Transactions on Circuits and Systems for Video Technology (IEEE-TCSVT)	2024-25
Journal of Open Source Software (JOSS)	2024
International Conference on Biomedical and Health Informatics (IEEE-EMBS BHI)	2024-25
British Machine Vision Conference (BMVC)	2020
Winter Conference on Applications of Computer Vision (WACV)	2021-23
European Conference on Computer Vision (ECCV)	2022
IET Biometrics	2019-25

Technical Skills

Languages: PyTorch (Geometric, Lightning), R, Keras, Python, SQL, Java, C/C++
Courses: Causal Inference, Machine Learning, Computer Vision, Object Oriented Programming, Multimodal Neuroimaging, Data Mining, Information Retrieval, Artificial Intelligence, Probability and Statistics
MOOCs: Data Science Specialization